AI 101 for Mechanical Engineers · Module 13

Tokens, context and embeddings

Lesson 13.2

What you will learn

> Distinguish tokens, context windows and embeddings, explain how context disambiguates technical terms, and identify the limits of long context and vector similarity.

Lesson overview

Module 13 · Lesson 2

Tokens, context and embeddings

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1 Tokens are model-specific units

"Turbocharger efficiency: 0.72"

↓ **tokenizer**

`Turbo` `charger` `␣efficiency` `:` `␣0` `.` `72`

Illustrative split only; each model has its own rules.

Here, 2 words and a number became 7 tokens. A word count does not give the exact token count.

2 Context: what the model can use

context window

question | units, system | supplied manual extract

datasheet not supplied

Information left outside cannot be assumed to have been consulted.

A larger window holds more, but does not guarantee perfect recall or correct use of every detail.

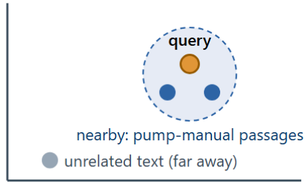
3 State the system to remove ambiguity

"cell"

PV cell	**Battery cell**	**Spreadsheet cell**
solar module	energy storage	data table

Say which system you mean, with units and assumptions. Attention combines this context in the computation; it does not check the physics.

4 Embeddings and similarity



An embedding is a numerical vector for an item: a word, or a whole passage. Nearby vectors help **retrieve** related passages. Being near does not prove that a passage supports the claim; read it.

5 Each mechanism has a limit

Mechanism	What it does	What it does not do
Tokenization	Splits input into units	Match word counts
Context window	Bounds the available information	Guarantee correct use
Embedding	Represents items as vectors	Establish meaning or truth
Similarity	Helps retrieve related text	Verify an engineering claim

Key takeaway

Supply the system, units and assumptions, then verify the answer.

Tokens, context and similarity help the model work; none of them checks the engineering.

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Tokenization depends on the model



- A token may be a word fragment or punctuation [72]
- One word may produce several tokens
- Word counts do not give exact token counts

Two separate questions about context

Was it supplied?

- Context window: finite available content
- An omitted datasheet cannot be assumed consulted

Was it used accurately?

- Larger window: no guarantee of perfect recall [73]
- No guarantee of factual correctness [73]

Long inputs or responses do not show units and assumptions were handled correctly.

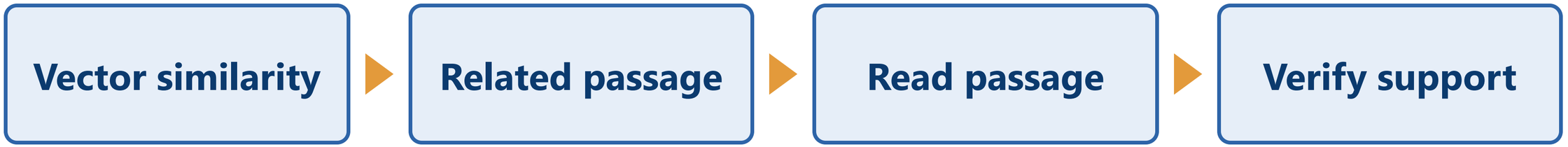
Context helps disambiguate “cell”

- Photovoltaic cell, battery cell or spreadsheet cell?
- State the system, units and assumptions
- Do not expect missing assumptions to be inferred correctly
- Attention combines representations [63]; not proof of physical interpretation

An embedding is a numerical vector

- Vectors can be learned for words [74]
- Retrieval systems can use vectors for whole passages [76]
- Vectors allow numerical comparison between items
- Tokens, context windows and embeddings are different parts

Similarity retrieves; reading verifies



- Proximity alone does not establish factual correctness
- It does not validate a real-equipment safety specification

Each function has a limit

Component	Function	Limit
Tokenization	Supplies encoded units	Word counts do not give token counts
Context window	Bounds available information	Capacity does not guarantee accurate use
Embeddings and similarity	Numerical vectors; assist retrieval	Relatedness does not verify claims

> Independent verification of the resulting claim remains necessary.

CHECK YOUR UNDERSTANDING

Why is counting words not a reliable exact token count?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Why is counting words not a reliable exact token count?

> Tokens may be word fragments, punctuation or other encoded units, and tokenization depends on the model. A word count does not identify exact token boundaries.

CHECK YOUR UNDERSTANDING

If a retrieved manual passage has a nearby embedding vector, has the engineering claim been verified?

Answer before moving on.

CHECK YOUR UNDERSTANDING

If a retrieved manual passage has a nearby embedding vector, has the engineering claim been verified?

No. Similarity indicates relatedness and helps retrieval. Check whether the actual passage supports the claim. Long context does not guarantee correct use.

Key takeaways

- Tokens are model-specific; word counts do not give exact token counts
- Larger context does not guarantee accurate use
- State the system, units and assumptions to reduce ambiguity
- Embeddings are numerical; similarity can help retrieve passages
- Relatedness and context do not verify engineering claims

References

[63] A. Vaswani et al., “Attention is all you need,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 30, 2017, pp. 5998–6008.

[72] R. Sennrich, B. Haddow, and A. Birch, “Neural machine translation of rare words with subword units,” in Proc. 54th Annu. Meeting Assoc. Comput. Linguistics (ACL), 2016, pp. 1715–1725. doi: 10.18653/v1/P16-1162

[73] N. F. Liu et al., “Lost in the middle: How language models use long contexts,” Trans. Assoc. Comput. Linguistics, vol. 12, pp. 157–173, 2024. doi: 10.1162/tacl_a_00638

[74] T. Mikolov, K. Chen, G. Corrado, and J. Dean, “Efficient estimation of word representations in vector space,” 2013, arXiv:1301.3781. doi: 10.48550/arXiv.1301.3781

[76] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.

Numbers match the course reference list. The written lesson is the authoritative text.